PROMPT ENGINEERING

20 Hands-on Activities: From Prompt Technique to Practical Output

A mixed-audience activity workbook for IT professionals, freshers, managers, educators, and non-technical participants.

How to run the activities

1. Assign one activity to each individual or small team.

2. Participants select and combine the listed prompt categories; they do not need to use every technique at once.

3. Replace example details with their own scenario, while avoiding confidential or personal information.

4. Review the output against the final verification step before presenting it.

Facilitator tip: Ask each participant to show the original prompt, the improved prompt, the AI output, and one change they made after checking quality. This makes the mapping between prompt technique and result visible.

Activity Mapping Table — Activities 1–5

Activity	Prompt category to use	Solution steps
1. Create a professional resume	Role-based; Context-rich; Audience-adapted; Output-format; Constraint-based	1. Provide education, skills, projects, experience, and target role. 2. Ask AI to act as a resume coach and identify relevant strengths. 3. Request a one- or two-page ATS-friendly resume with clear headings. 4. Ask it to use action verbs and measurable results only when supplied. 5. Check every claim, dates, spelling, and formatting; remove invented details.
2. Tailor a resume to a job description	Information extraction; Comparative; Context-rich; Constraint-based; Verification	1. Paste the resume and job description. 2. Extract required skills and responsibilities from the job description. 3. Compare them with evidence in the resume and show matches and gaps in a table. 4. Rewrite relevant bullets without adding skills or experience the candidate does not have. 5. Verify that every revised claim is accurate.
3. Write a cover letter or job application email	Role-based; Context-rich; Audience-adapted; Rewrite; Constraint-based	1. Provide the role, company, job description, and two or three genuine achievements. 2. Ask AI to draft a professional, specific letter. 3. Limit it to the requested length and avoid generic claims. 4. Request a subject line and a concise email version. 5. Personalize the final wording and verify names and facts.
4. Create a business or project report	Context-rich; Task-specific; Summarization; Output-format; Grounded/source-based	1. Provide the report purpose, audience, source data, and reporting period. 2. Ask AI to organize the report into executive summary, findings, evidence, risks, and recommendations. 3. Require each finding to link to supplied evidence and label unknowns. 4. Request tables or charts only when the data supports them. 5. Check calculations, source references, and conclusions before sharing.
5. Turn meeting notes into action items	Information extraction; Structured output; Summarization; Verification	1. Paste meeting notes or a transcript, removing unnecessary sensitive information. 2. Ask AI to extract decisions, action items, owners, due dates, and open questions. 3. Request a table and use “Not stated” for missing owners or deadlines. 4. Ask for a short meeting summary and follow-up email. 5. Confirm the action list with attendees; do not let AI invent commitments.

Activity Mapping Table — Activities 6–10

Activity	Prompt category to use	Solution steps
6. Draft a professional email for a difficult situation	Role-based; Audience-adapted; Rewrite; Negative/exclusion; Constraint-based	1. Describe the situation, recipient, desired outcome, and facts that must be included. 2. Ask for a calm, respectful draft with a clear request or next step. 3. Exclude blame, threats, unsupported claims, and unnecessary personal details. 4. Ask for a warmer and a more direct alternative. 5. Review tone and factual accuracy before sending.
7. Build a presentation from a topic or report	Task decomposition; Audience-adapted; Output-format; Constraint-based; Iterative refinement	1. Specify audience, presentation duration, goal, and source material. 2. Ask for a slide-by-slide outline with one key message per slide. 3. Request suggested visuals, examples, and speaker notes. 4. Ask AI to check flow, repetition, and timing. 5. Build the slides in a presentation tool and verify all facts and visuals.
8. Analyze a spreadsheet or CSV	Task-specific; Step-by-step; Reasoning/analysis; Structured output; Verification	1. Describe the columns, business question, and meaning of each field; upload data only where permitted. 2. Ask AI to propose data-quality checks before analysis. 3. Request suitable summaries, comparisons, and charts with assumptions stated. 4. If using code, run it on a copy and inspect the output. 5. Recalculate key figures and avoid claiming causation from simple correlations.
9. Generate a Python solution and test it	Code-generation; Context-rich; Constraint-based; Step-by-step; Self-correction	1. State the task, Python version, libraries, input, and expected output. 2. Ask for readable code with comments and error handling. 3. Request sample inputs, expected results, and unit tests. 4. Run the code in a safe environment; share any traceback for debugging. 5. Test edge cases and review security before using it in production.
10. Create a beginner learning plan	Audience-adapted; Context-rich; Task decomposition; Constraint-based; Output-format	1. Specify the learner’s current level, goal, available hours, and deadline. 2. Ask for a week-by-week or day-by-day roadmap. 3. Include short learning tasks, practice exercises, and checkpoints. 4. Request a realistic schedule with buffer time. 5. Adjust the plan after a trial week based on progress.

Activity Mapping Table — Activities 11–15

Activity	Prompt category to use	Solution steps
11. Create a customer FAQ from policy documents	Grounded/source-based; RAG; Information extraction; Output-format; Hallucination-control	1. Use approved policy or product documents as the source. 2. Ask AI to identify common customer questions and locate supporting passages. 3. Generate concise answers with section or page references. 4. Instruct it to say “Not found in the source” when evidence is missing. 5. Have a responsible staff member approve the FAQ before publication.
12. Draft a standard operating procedure (SOP)	Role-based; Task decomposition; Step-by-step; Output-format; Verification	1. Select a repeatable process and provide the actual steps, roles, and tools. 2. Ask AI to structure purpose, scope, prerequisites, procedure, exceptions, and escalation. 3. Request numbered steps and a checklist for completion. 4. Mark unknown steps as questions instead of inventing them. 5. Ask the process owner to test and approve the SOP.
13. Create a social media campaign	Role-based; Audience-adapted; Creative generation; Constraint-based; Iterative refinement	1. Define the audience, goal, platform, topic, brand tone, and campaign dates. 2. Ask for several content angles and select the strongest. 3. Generate a calendar with post copy, call to action, and suggested visual. 4. Apply platform-specific length and brand constraints. 5. Fact-check claims, obtain image permissions, and review before posting.
14. Summarize a research paper	Grounded/source-based; Summarization; Information extraction; Comparative; Verification	1. Provide the paper or its text and state the research question you care about. 2. Ask for the objective, method, dataset, results, limitations, and conclusion. 3. Require evidence from the paper and distinguish author claims from your interpretation. 4. Compare with a second paper only if both sources are supplied or reliably retrieved. 5. Verify quotations, figures, and citations in the original paper.
15. Prepare a manager’s decision memo	Decision-support; Comparative; Context-rich; Reasoning/analysis; Output-format	1. State the decision to make, available options, budget, deadline, and priorities. 2. Ask AI to compare options using the same criteria. 3. Request benefits, costs, risks, assumptions, and missing information. 4. Ask for a recommendation conditional on the stated priorities. 5. Validate figures and involve relevant stakeholders before deciding.

Activity Mapping Table — Activities 16–20

Activity	Prompt category to use	Solution steps
16. Plan a workshop or team event	Task decomposition; Constraint-based; Audience-adapted; Output-format; Verification	1. Specify audience size, objective, duration, venue or online format, and budget. 2. Ask for an agenda with timings, activities, materials, and breaks. 3. Request a preparation checklist and contingency plan. 4. Check total time, resource needs, accessibility, and budget arithmetic. 5. Assign owners and confirm logistics with the team.
17. Analyze survey feedback	Information extraction; Classification; Summarization; Reasoning/analysis; Verification	1. Provide anonymized responses and the survey question. 2. Ask AI to group responses into themes using clearly defined labels. 3. Request counts or proportions only when they can be calculated from the data. 4. Include representative short examples without exposing identities. 5. Manually inspect ambiguous responses and report sampling limitations.
18. Prepare a client discovery questionnaire	Role-based; Open-ended; Probing/follow-up; Task decomposition; Output-format	1. Describe the client, project type, and what is not yet known. 2. Ask AI to draft open questions about goals, users, data, constraints, success measures, and risks. 3. Add probing follow-ups for vague answers. 4. Organize questions from business context to technical details. 5. Remove leading questions and tailor the list to the meeting time.
19. Plan a personal or business budget	Context-rich; Structured output; Task-specific; Decision-support; Verification	1. Enter realistic income, expense categories, timeframe, and financial goal; omit account numbers and secrets. 2. Ask AI to organize the figures into a budget table. 3. Request totals, differences, and possible savings areas based only on supplied numbers. 4. Ask for conservative scenarios and identify assumptions. 5. Verify arithmetic and treat the output as planning support, not guaranteed financial advice.
20. Design a small AI assistant or agent workflow	Combined prompt framework; Tool-use; Agent planning/replanning; Memory-aware; Guardrail/safety	1. Define the user, goal, inputs, expected output, and tasks the assistant must not perform. 2. Map the workflow into steps: understand request, retrieve or use an approved tool, check result, respond. 3. Specify when to ask a clarifying question and how to handle missing evidence or tool failure. 4. Define privacy, permissions, and confirmation requirements for actions with consequences. 5. Test normal, ambiguous, and adversarial cases; record failures and improve the prompt.

Quick Debrief and Assessment

Review area	Discussion question
Prompt technique	Which technique changed the quality of the result most, and why?
Evidence	Which facts, figures, or claims did you independently verify?
Iteration	What did you change after the first AI response?
Limitations	What did the AI not know, and how did you handle missing information?
Responsible use	Did you protect confidential information and review the output before using it?

Suggested scoring (10 points): Prompt clarity (2), suitable category mapping (2), usefulness of output (2), verification and accuracy (2), responsible use and explanation (2).